

Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

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Agricultura

Maximize

$$Z = 10000.00x_1 + 3000.00x_2$$

Subject to

$$\begin{aligned}1.00x_1 + 1.00x_2 &\leq 7.00 \\10.00x_1 + 4.00x_2 &\leq 40.00 \\0.00x_1 + 1.00x_2 &\geq 3.00\end{aligned}$$

Simplex Table

	Z	x_1	x_2	s_1	s_2	e_1	a_1	b
	1	-10000	-3000	0	0	0	M	0
	0	1	1	1	0	0	0	7
	0	10	4	0	1	0	0	40
	0	0	1	0	0	-1	1	3

Results

	Z	x_1	x_2	s_1	s_2	e_1	a_1	b
	1	0	0	0	1000	1000	0.999M	37000
	0	0	0	1	-0.1	0.6	-0.6	1.2
	0	1	0	0	0.1	0.4	-0.4	2.8
	0	0	1	0	0	-1	1	3

Objective Value

$$Z = 37000.00$$

Variables

$$x_1 = 2.80$$

$$x_2 = 3.00$$

Slack or Surplus

$$s_1 = 1.20$$

$$s_2 = 0.00$$

$$s_3 = 0.00$$